

PSTAT 131 Final Project

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Copied from instructions: Predicting voter behavior is complicated for many reasons despite the tremendous effort in collecting, cleaning, analyzing, and understanding many available datasets. For this final project, we will visualize and analyze the 2024 U.S. presidential election. In addition, we will combine the United States county-level census data with the election data. Our target would then be building and selecting classification models (among many predictive models that we've covered in this quarter) to predict the election winner.

Data

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
## read election data
```

```
election_raw <- read_csv("election_raw.csv", col_names = TRUE)
```

```
Rows: 3160 Columns: 5
```

```
-- Column specification -----
Delimiter: ","
chr (2): State, County
```

```
dbl (3): FIPS, votes_gop, votes_dem
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
## read census data
census <- read_csv("census.csv")
```

```
Rows: 3222 Columns: 14
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (3): County, State, Urban-Rural
```

```
dbl (11): State FIPS Code, County FIPS Code, FIPS, Total Population, Male Po...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
1. election_raw
```

```
print("dimension of election_raw:")
```

```
[1] "dimension of election_raw:"
```

```
dim(election_raw)
```

```
[1] 3160    5
```

```
print("missing value count:")
```

```
[1] "missing value count:"
```

```
sum(is.na(election_raw))
```

```
[1] 0
```

```
print("distinct 'state' values:")
```

```
[1] "distinct 'state' values:"
```

```
n_distinct(election_raw$State)
```

```
[1] 51
```

2. census

```
print("dimension of census dataframe:")
```

```
[1] "dimension of census dataframe:"
```

```
dim(census)
```

```
[1] 3222  14
```

```
print("missing value count:")
```

```
[1] "missing value count:"
```

```
sum(is.na(census))
```

```
[1] 0
```

```
print("distinct 'county' values:")
```

```
[1] "distinct 'county' values:"
```

```
n_distinct(census$County)
```

```
[1] 1960
```

We computed 1960 distinct county values in the census dataframe. However, election_raw reports 3160 counties in election_raw. This is because many states share county

Data Wrangling

3. Aggregated dataset

```
library(dplyr)
election_state <- election_raw%>%group_by(State)%>%summarise(total_gop=sum(votes_gop), total.
```

4. State and County Winners

```
county_winner <- election_raw%>%
  group_by(State,County)%>%
  summarise(
    total_gop=sum(votes_gop),
    total_dem=sum(votes_dem),
    winner=ifelse(total_gop>total_dem, "gop","dem"),
    FIPS = FIPS,
    .groups = 'drop'
  )
state_winner <- election_raw%>%
  group_by(State)%>%
  summarise(
    total_gop=sum(votes_gop),
    total_dem=sum(votes_dem),
    winner=ifelse(total_gop>total_dem, "gop","dem"),
    .groups = 'drop'
  )
```

Visualization

Copied from instructions: Visualization is crucial for gaining insight and intuition during data mining. We will map our data onto maps. In this part, we will work towards building state/county-level red/blue map plots that are commonly shown on media coverage.

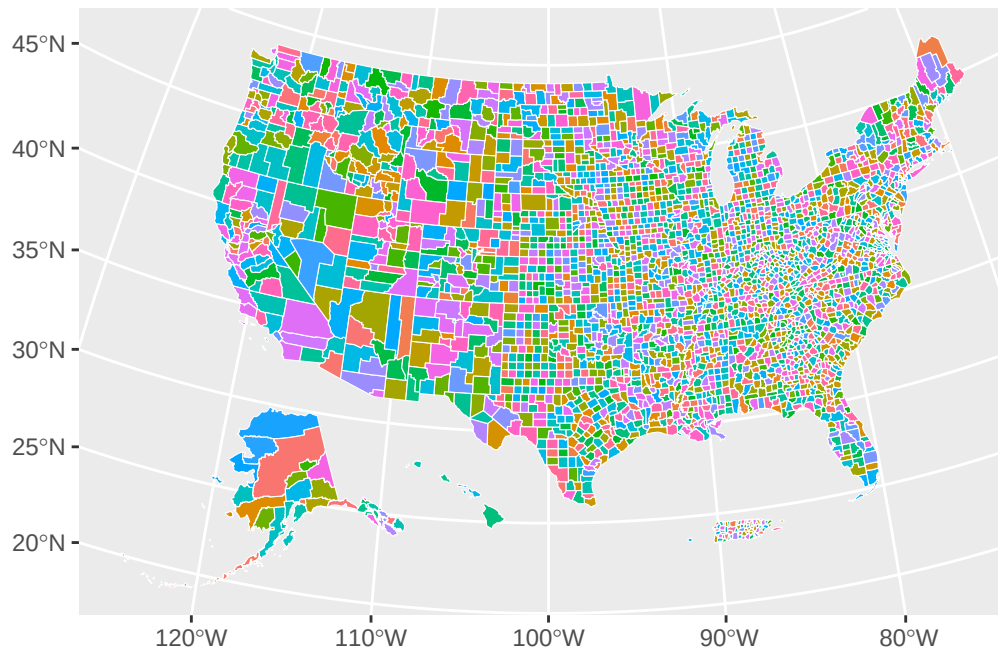
5. county map

```
library(usmap)
```

Warning: package 'usmap' was built under R version 4.5.3

```
states <- usmap::us_map(regions="counties")

ggplot(states) +
  geom_sf(aes(fill=county), color = "white") +
  coord_sf() +
  guides(fill = "none")
```

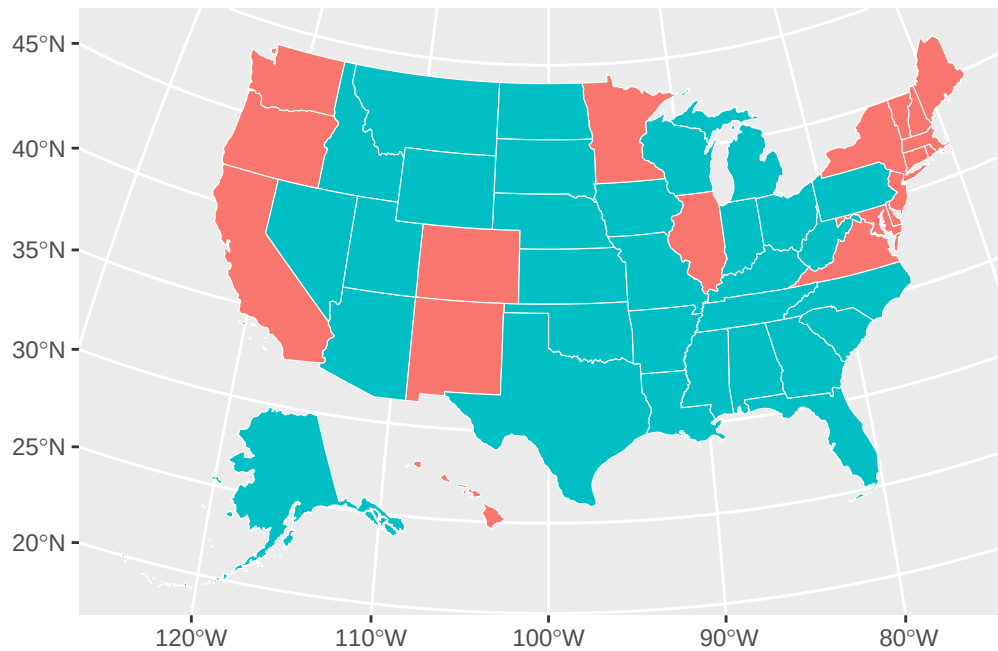


6. state map

```
# state-level plotting data
states <- states%>%
  filter(abbr != "PR")%>%
  group_by(full)%>%
  summarise(
    .groups = 'drop'
  )

states <- left_join(states, state_winner, by=c("full"="State"))
```

```
ggplot(states) +
  geom_sf(aes(fill=winner), color = "white") +
  coord_sf() +
  guides(fill = "none")
```

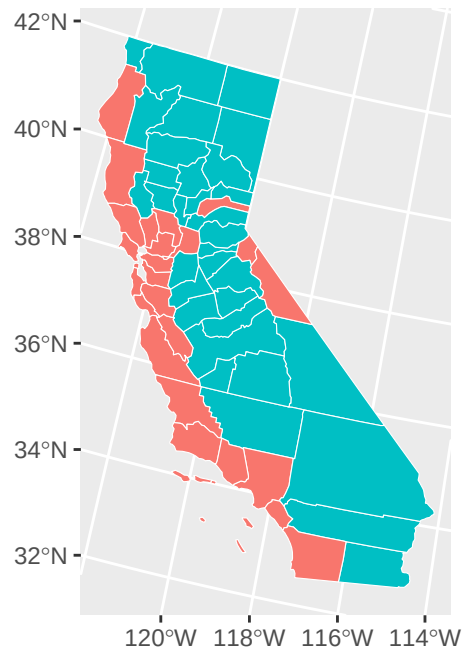


7. California

```
ca_counties <- usmap::us_map(regions="counties", include = "CA")
ca_winner <- county_winner%>%
  filter(State == "California")

ca_counties <- left_join(ca_counties, ca_winner, by=c("county"="County"))

ggplot(ca_counties) +
  geom_sf(aes(fill=winner), color = "white") +
  coord_sf() +
  guides(fill = "none")
```

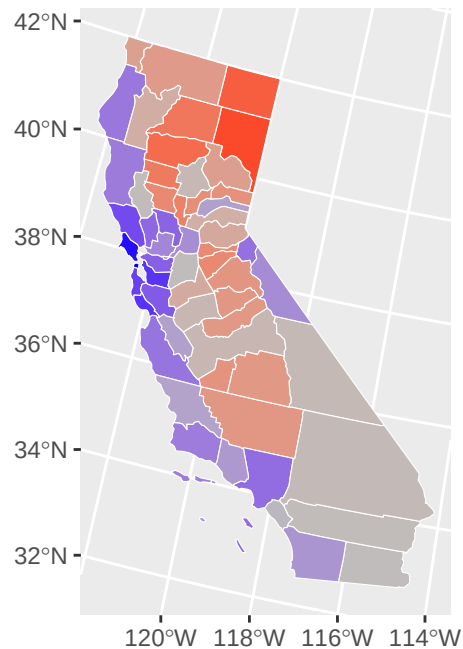


8. Open-ended

Vote share gradient for Democratic vs Republican.

```
ca_county_data <- ca_winner%>%
  mutate(total_votes = total_gop+total_dem,
         vote_share = total_dem/total_votes)

ca_counties <- left_join(ca_counties, ca_county_data, by = c("county"="County"))
ggplot(ca_counties) +
  geom_sf(aes(fill=vote_share), color = "white") +
  coord_sf() +
  scale_fill_gradient2(
    low = "red",
    mid = 'grey',
    high = 'blue',
    midpoint = 0.5,
    name = "Vote Share",
    labels = scales::percent
  ) +
  guides(fill = "none")
```



9. Clean county-level census data

```
census_clean <- census%>%
  na.omit() %>%
  mutate(
    pct_male = `Male Population`/`Total Population`,
    pct_female = `Female Population`/`Total Population`,
    pct_whitealone = `White Alone`/`Total Population`,
    pct_black = `Black or African American Alone`/`Total Population`,
    pct_hispanic = `Hispanic or Latino`/`Total Population`,
    Urban = factor(ifelse(`Urban-Rural`=="Urban", 1,0))
  ) %>%
  select(
    County,
    State,
    Urban,
    `Median Household Income`,
    pct_male,
    pct_female,
    pct_whitealone,
    pct_black,
    pct_hispanic
```



```
)
head(census_clean)
```

```
# A tibble: 6 x 9
  County State Urban Median Household Inc~1 pct_male pct_female pct_whitealone
  <chr>   <chr> <fct>          <dbl>    <dbl>    <dbl>    <dbl>
1 Autauga~ Alab~ 1             69841    0.484    0.516    0.736
2 Baldwin~ Alab~ 1             75019    0.489    0.511    0.828
3 Barbour~ Alab~ 0             44290    0.521    0.479    0.440
4 Bibb Co~ Alab~ 0             51215    0.534    0.466    0.751
5 Blount ~ Alab~ 1             61096    0.505    0.495    0.895
6 Bullock~ Alab~ 0             36723    0.547    0.453    0.229
# i abbreviated name: 1: `Median Household Income`
# i 2 more variables: pct_black <dbl>, pct_hispanic <dbl>
```

10. Run PCA for the cleaned county level census data

```
pca_data <- census_clean %>%
  select(-County, -State)

pca_data$Urban <- as.numeric(as.character(pca_data$Urban))

pca_result <- prcomp(
  pca_data,
  center = TRUE,
  scale. = TRUE,
)

biplot(pca_result)
```


The features were centered and scaled before running PCA because PCA is sensitive to variance of the input features. So features with large raw magnitudes would dominate if we don't normalize them.

The three highest absolute values for PC1 are pct_male, pct_female, and pct_whitealone. For PC1, pct_male and white alone has an opposite sign from the others. Negative sign indicates a feature's favorability towards the democratic candidate, and positive indicates favorability towards the republican. This means male and white alone are inversely correlated with everything else. This makes sense since the PC1 values of male and female pct are exactly the opposite sign from each other. We addressed the issue of Urban-Rural being qualitative by converting it to binary classification earlier.

11.

```
vars <- (pca_result$sdev)^2/sum((pca_result$sdev)^2)
cum_var <- cumsum(vars)
cum_var
```

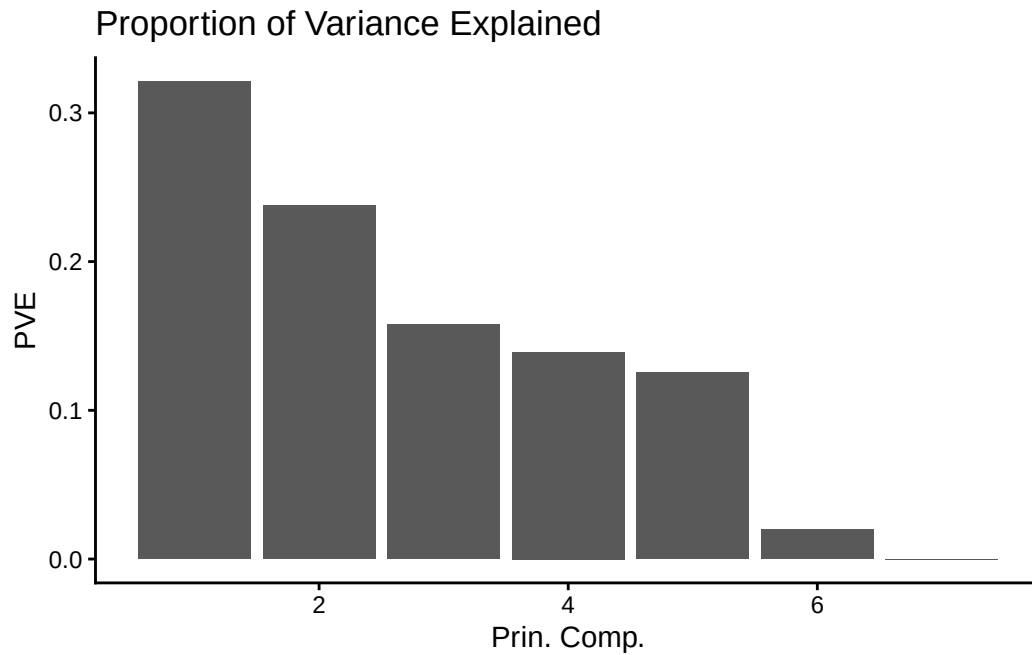
```
[1] 0.3210266 0.5584560 0.7160544 0.8552634 0.9804067 1.0000000 1.0000000
```

We need at least 5 PCs to capture 90% of the variance for the analysis.

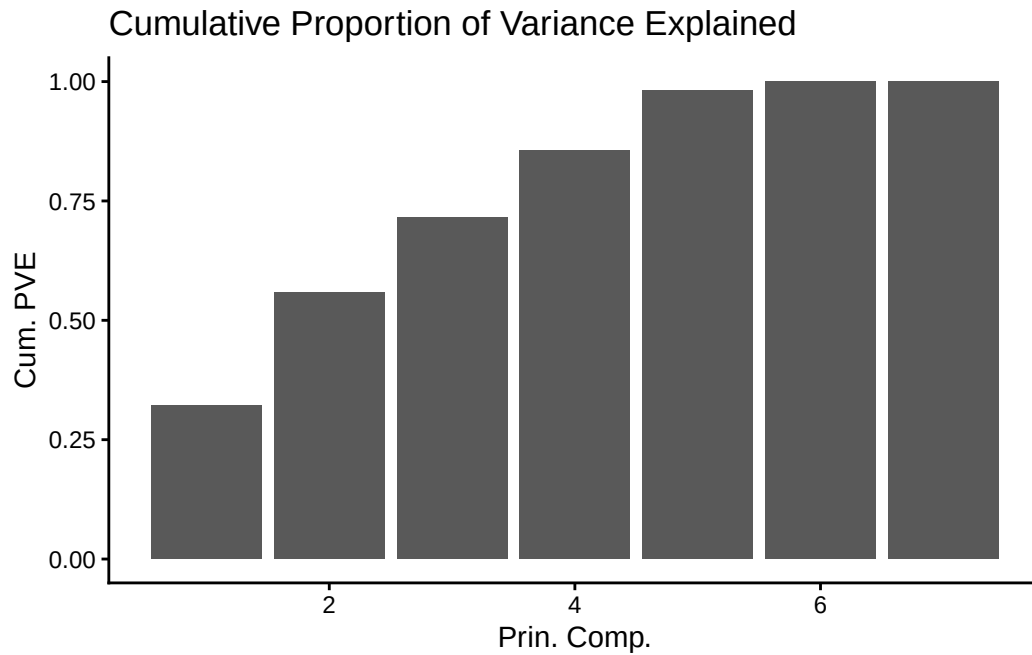
Plots:

```
pve <- data.frame(
  PC = 1:length(vars),
  PVE = vars,
  CumPVE = cum_var
)

ggplot(pve, aes(x=PC, y=PVE))+
  geom_col() +
  labs(
    title = "Proportion of Variance Explained",
    x = "Prin. Comp.",
    y = "PVE"
  ) +
  theme_classic()
```



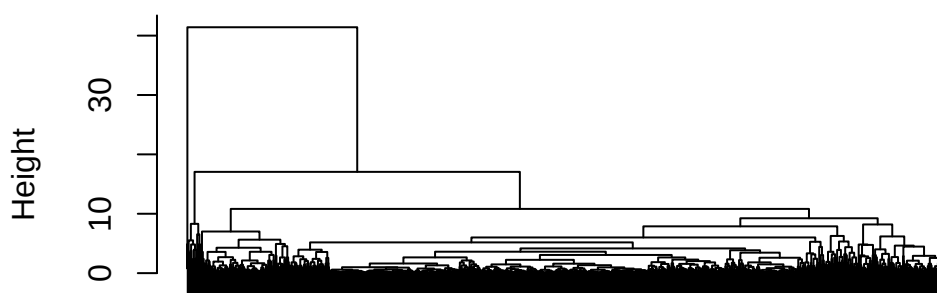
```
ggplot(pve, aes(x=PC, y=CumPVE))+  
  geom_col() +  
  labs(  
    title = "Cumulative Proportion of Variance Explained",  
    x = "Prin. Comp.",  
    y = "Cum. PVE"  
  ) +  
  theme_classic()
```



12. Clustering

```
cluster_data <- census_clean %>%  
  select(-State, -County)  
  
cluster_data$Urban <- as.numeric(as.character(cluster_data$Urban))  
  
cluster_scaled <- scale(cluster_data)  
  
hc_original <- hclust(dist(cluster_scaled), method = 'complete')  
  
plot(  
  hc_original,  
  labels = FALSE,  
  main = "Hierarchical Clustering Using Census Variables",  
  xlab = " ",  
  sub = " "  
)
```

Hierarchical Clustering Using Census Variables



```
clusters_original <- cutree(hc_original, k = 10)

census_clusters_original <- census_clean %>%
  mutate(cluster_original = clusters_original)

santa_barbara_original <- census_clusters_original %>%
  filter(State == "California", County == "Santa Barbara County")

santa_barbara_original
```

```
# A tibble: 1 x 10
  County State Urban Median Household Inc~1 pct_male pct_female pct_whitealone
  <chr>   <chr> <fct>          <dbl>    <dbl>      <dbl>      <dbl>
1 Santa B~ Cali~ 1              95977    0.502    0.498    0.543
# i abbreviated name: 1: `Median Household Income`
# i 3 more variables: pct_black <dbl>, pct_hispanic <dbl>,
#   cluster_original <int>
```

```
pc_scaled <- scale(pc_county)

hc_pc <- hclust(dist(pc_scaled), method = "complete")
```

```
plot(
  hc_pc,
  labels = FALSE,
  main = "Hierarchical Clustering Using PC1 and PC2",
  xlab = " ",
  sub = " "
)
```

Hierarchical Clustering Using PC1 and PC2



```
clusters_pc <- cutree(hc_pc, k = 10)

census_clusters_pc <- census_clean %>%
  mutate(clusters_pc = clusters_pc)

head(census_clusters_pc)
```

A tibble: 6 x 10

	County	State	Urban	Median Household Inc~1	pct_male	pct_female	pct_whitealone
	<chr>	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	Autauga~	Alab~	1	69841	0.484	0.516	0.736
2	Baldwin~	Alab~	1	75019	0.489	0.511	0.828
3	Barbour~	Alab~	0	44290	0.521	0.479	0.440
4	Bibb Co~	Alab~	0	51215	0.534	0.466	0.751

```

5 Blount ~ Alab~ 1          61096    0.505    0.495    0.895
6 Bullock~ Alab~ 0          36723    0.547    0.453    0.229
# i abbreviated name: 1: `Median Household Income`
# i 3 more variables: pct_black <dbl>, pct_hispanic <dbl>, clusters_pc <int>

```

```

santa_barbara_pc <- census_clusters_pc %>%
  filter(State == "California", County == "Santa Barbara County")

santa_barbara_pc

```

```

# A tibble: 1 x 10
  County State Urban Median Household Inc~1 pct_male pct_female pct_whitealone
  <chr>   <chr> <fct>          <dbl>    <dbl>    <dbl>    <dbl>
1 Santa B~ Cali~ 1          95977    0.502    0.498    0.543
# i abbreviated name: 1: `Median Household Income`
# i 3 more variables: pct_black <dbl>, pct_hispanic <dbl>, clusters_pc <int>

```

```

sb_original_cluster <- santa_barbara_original$cluster_original

sb_original_counties <- census_clusters_original %>%
  filter(cluster_original == sb_original_cluster) %>%
  select(State, County, 'Median Household Income', Urban, pct_male, pct_female, pct_whitealone)

head(sb_original_counties, 15)

```

```

# A tibble: 15 x 9
  State County Median Household Inc~1 Urban pct_male pct_female pct_whitealone
  <chr>  <chr>          <dbl> <fct>    <dbl>    <dbl>    <dbl>
1 Alaba~ Autau~      69841 1      0.484    0.516    0.736
2 Alaba~ Baldw~      75019 1      0.489    0.511    0.828
3 Alaba~ Barbo~      44290 0      0.521    0.479    0.440
4 Alaba~ Bibb ~      51215 0      0.534    0.466    0.751
5 Alaba~ Bloun~      61096 1      0.505    0.495    0.895
6 Alaba~ Calho~      55826 1      0.483    0.517    0.710
7 Alaba~ Chero~      50769 0      0.493    0.507    0.913
8 Alaba~ Chilt~      61873 1      0.489    0.511    0.809
9 Alaba~ Clay ~      51852 0      0.482    0.518    0.805
10 Alaba~ Clebu~      53319 0      0.502    0.498    0.928
11 Alaba~ Coffe~      64672 1      0.497    0.503    0.715
12 Alaba~ Colbe~      56736 1      0.482    0.518    0.780
13 Alaba~ Coosa~      57063 0      0.512    0.488    0.646

```



```

14 Alaba~ Covin~          50886 1          0.483          0.517          0.832
15 Alaba~ Crens~          49040 0          0.484          0.516          0.706
# i abbreviated name: 1: `Median Household Income`
# i 2 more variables: pct_black <dbl>, pct_hispanic <dbl>

```

```

sb_pc_cluster <- santa_barbara_pc$clusters_pc

sb_pc_counties <- census_clusters_pc %>%
  filter(clusters_pc == sb_pc_cluster) %>%
  select(State, County, 'Median Household Income', Urban, pct_male, pct_female, pct_whitealone)

head(sb_pc_counties, 15)

```

```

# A tibble: 15 x 9
  State County Median Household Inc~1 Urban pct_male pct_female pct_whitealone
<chr> <chr>          <dbl> <fct>    <dbl>    <dbl>    <dbl>
1 Alaba~ Barbo~          44290 0          0.521    0.479    0.440
2 Alaba~ Bullo~          36723 0          0.547    0.453    0.229
3 Alaska Bethe~          70616 0          0.535    0.465    0.0875
4 Alaska Dilli~          74250 0          0.516    0.484    0.153
5 Alaska Kusil~          48233 0          0.535    0.465    0.0319
6 Alaska Nome ~          76643 0          0.523    0.477    0.141
7 Alaska North~          81298 0          0.528    0.472    0.104
8 Alaska Yukon~          52642 0          0.533    0.467    0.227
9 Arizo~ Green~          75239 0          0.523    0.477    0.647
10 Arizo~ Yuma ~          60417 1          0.516    0.484    0.484
11 Arkan~ Dalla~          44694 0          0.502    0.498    0.529
12 Arkan~ Lafay~          37237 0          0.515    0.485    0.577
13 Arkan~ Lee C~          36860 0          0.560    0.440    0.400
14 Arkan~ St. F~          41784 0          0.555    0.445    0.411
15 Calif~ Colus~          75149 0          0.516    0.484    0.512
# i abbreviated name: 1: `Median Household Income`
# i 2 more variables: pct_black <dbl>, pct_hispanic <dbl>

```

The PCA based clustering placed Santa Barbara County in a more appropriate cluster. the original clustering grouped Santa Barbara with over 2000 counties while the PCA based clustering produced a smaller and more meaningful group. This most likely happened because PCA reduces noise and multicollinearity by focusing on the main sources of variation in the data.

Classification

```
# we move all state and county names into lower-case
# we further remove suffixes of "county"
tmpwinner <- county_winner %>% ungroup %>%
  mutate_at(vars(State, County), tolower) %>%
  mutate(County = gsub(" county", "", County))

# we move all state and county names into lower-case
# we further remove suffixes of "county"
tmpcensus <- census_clean %>% mutate_at(vars(State, County), tolower) %>%
  mutate(County = gsub(" county", "", County))

# we join the two datasets
election_cl <- tmpwinner %>%
  left_join(tmpcensus, by = c("State" = "State", "County" = "County")) %>%
  na.omit

## save meta information
election_meta <- election_cl %>% select(c(State, FIPS, County, total_gop, total_dem))

## save predictors and class labels
election_cl = election_cl %>% select(-c(State, FIPS, County, total_gop, total_dem, pct_femal
set.seed(10)
n <- nrow(election_cl)
idx.tr <- sample.int(n, 0.8*n)
election.tr <- election_cl[idx.tr, ]
election.te <- election_cl[-idx.tr, ]
set.seed(20)
nfold <- 10
folds <- sample(cut(1:nrow(election.tr), breaks=nfold, labels=FALSE))
calc_error_rate = function(predicted.value, true.value){
  return(mean(true.value!=predicted.value))
}
records = matrix(NA, nrow=3, ncol=2)
colnames(records) = c("train.error", "test.error")
rownames(records) = c("tree", "logistic", "lasso")
```

13. decision tree

```
names(election.tr)[names(election.tr) == "Median Household Income"] <-
  "Median_Household_Income"

names(election.te)[names(election.te) == "Median Household Income"] <-
  "Median_Household_Income"
election.tr$winner <- factor(election.tr$winner)
election.te$winner <- factor(election.te$winner)
str(election.tr$winner)
```

Factor w/ 2 levels "dem","gop": 2 2 2 2 2 2 1 2 2 2 ...

```
library(tree)
```

Warning: package 'tree' was built under R version 4.5.3

```
tree_fit <- tree(
  winner ~ .,
  data = as.data.frame(election.tr)
)
cv_output <- cv.tree(
  tree_fit,
  FUN = prune.misclass,
  rand = folds
)

best_size <- cv_output$size[which.min(cv_output$dev)]

pruned_tree <- prune.misclass(
  tree_fit,
  best = best_size
)

train_pred <- predict(
  pruned_tree,
  election.tr,
  type = "class"
)

test_pred <- predict(
```

```

pruned_tree,
election.te,
type = "class"
)

train_error <- mean(train_pred != election.tr$winner)
test_error  <- mean(test_pred  != election.te$winner)

records["tree", "train.error"] <- train_error
records["tree", "test.error"]  <- test_error

```

store

```
records
```

	train.error	test.error
tree	0.1004419	0.117175
logistic	NA	NA
lasso	NA	NA

interpretation

```
pruned_tree
```

```

node), split, n, deviance, yval, (yprob)
      * denotes terminal node

```

```

1) root 2489 2029.00 gop ( 0.14142 0.85858 )
 2) pct_whitealone < 0.499699 205 260.50 dem ( 0.66829 0.33171 )
 4) pct_black < 0.543108 150 205.80 dem ( 0.56000 0.44000 )
    8) Median_Household_Income < 76423.5 111 152.40 gop ( 0.44144 0.55856 ) *
    9) Median_Household_Income > 76423.5 39 25.79 dem ( 0.89744 0.10256 ) *
 5) pct_black > 0.543108 55 17.18 dem ( 0.96364 0.03636 ) *
 3) pct_whitealone > 0.499699 2284 1425.00 gop ( 0.09413 0.90587 )
 6) Median_Household_Income < 83618 2005 841.10 gop ( 0.05387 0.94613 )
   12) Urban: 0 1110 167.40 gop ( 0.01441 0.98559 ) *
   13) Urban: 1 895 592.80 gop ( 0.10279 0.89721 )
      26) pct_male < 0.492636 297 287.60 gop ( 0.18855 0.81145 )
         52) Median_Household_Income < 58849 107 27.36 gop ( 0.02804 0.97196 ) *
         53) Median_Household_Income > 58849 190 224.90 gop ( 0.27895 0.72105 ) *

```

```

27) pct_male > 0.492636 598 272.10 gop ( 0.06020 0.93980 ) *
7) Median_Household_Income > 83618 279 371.50 gop ( 0.38351 0.61649 )
14) pct_whitealone < 0.709607 86 114.50 dem ( 0.61628 0.38372 ) *
15) pct_whitealone > 0.709607 193 228.80 gop ( 0.27979 0.72021 ) *

```

```

library(rpart)
library(rpart.plot)

```

Warning: package 'rpart.plot' was built under R version 4.5.3

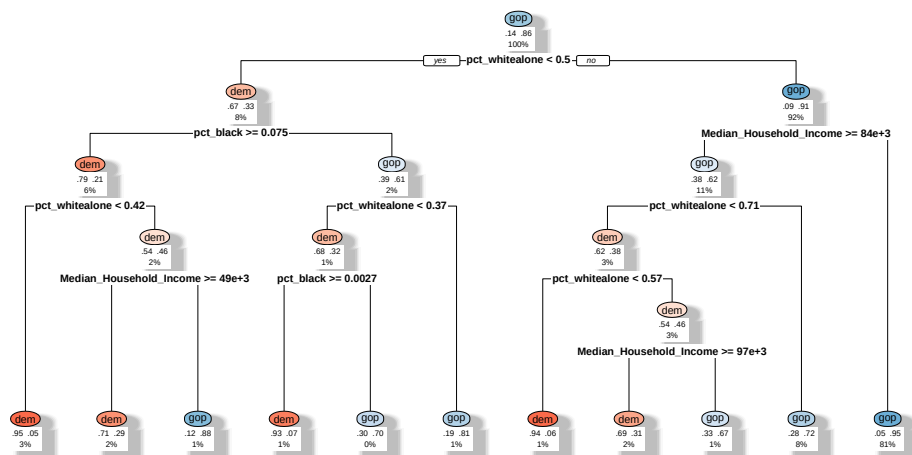
```

rpart_fit <- rpart(winner ~ .,
  data = election.tr,
  method = "class",
  control = rpart.control(cp = 0.01))

rpart.plot(rpart_fit,
  type = 2,
  extra = 104,
  under = TRUE,
  faclen = 0,
  box.palette = "RdBu",
  shadow.col = "gray",
  main = "Predicting Election Winners")

```

Predicting Election Winners



pct_white alone is at the top of the decision tree, meaning it is the top predictor. One of the most interesting things we found was that if the white percentage is greater than 50%, and the household income is greater than 83618, then the county is likely to vote democratic if it has less than 70% white people. This is basically describing California. Also, if the black percentage is greater than 54%, then the county will vote democratic without any conditions. Also, if it is less than that, having LESS than \$76423 household income means a county will vote GOP.

14. logistic regression

```
log_fit <- glm(
  winner ~ .,
  data = election.tr,
  family = "binomial"
)
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
summary(log_fit)
```

Call:

```
glm(formula = winner ~ ., family = "binomial", data = election.tr)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.961e+00	1.762e+00	-2.247	0.0246 *
Urban1	-7.497e-01	1.763e-01	-4.252	2.12e-05 ***
Median_Household_Income	-4.347e-05	3.999e-06	-10.872	< 2e-16 ***
pct_male	7.987e+00	3.323e+00	2.404	0.0162 *
pct_whitealone	7.459e+00	6.719e-01	11.102	< 2e-16 ***
pct_black	-1.470e+00	7.080e-01	-2.076	0.0379 *
pct_hispanic	1.283e+00	6.165e-01	2.081	0.0374 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2028.7 on 2488 degrees of freedom

Residual deviance: 1330.4 on 2482 degrees of freedom

AIC: 1344.4

Number of Fisher Scoring iterations: 10

In the training set, all of the variables are significant at 0.05. Urban, Income, and White are the most significant. White has the highest coefficient out of the most significant variables.

predictions (probabilities of GOP)

```
train_prob <- predict(
  log_fit,
  newdata = election.tr,
  type = "response"
)
test_prob <- predict(
  log_fit,
  newdata = election.te,
  type = "response"
)
logtrain_pred <- ifelse(train_prob>0.5, "gop", "dem")
logtest_pred <- ifelse(test_prob>0.5, "gop", "dem")

logtrain_pred <- factor(
  logtrain_pred,
  levels = levels(election.tr$winner)
)

logtest_pred <- factor(
  logtest_pred,
  levels = levels(election.te$winner)
)

logtrain_error <- mean(
  logtrain_pred != election.tr$winner
)

logtest_error <- mean(
  logtest_pred != election.te$winner
)
records["logistic", "train.error"] <- logtrain_error
records["logistic", "test.error"] <- logtest_error
```

```
exp(coef(log_fit)*0.01)
```

(Intercept)	Urban1	Median_Household_Income
0.9611665	0.9925314	0.9999996

pct_male	pct_whitealone	pct_black
1.0831424	1.0774450	0.9854116
pct_hispanic		
1.0129145		

A 1% increase in the white share of a county's population increases the odds of the county voting republican by 7.74%, keeping everything else constant. A 1% increase in the male share of a county's population increases the odds of the county voting republican by 8.31%, keeping everything else constant, but this is less significant. Probably because there is less variance in this group than in white population in different counties.

15.

```
library(glmnet)
```

Warning: package 'glmnet' was built under R version 4.5.2

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loaded glmnet 4.1-10

```
x_train <- model.matrix(winner ~ ., data = election.tr)[, -1]
y_train <- election.tr$winner

x_test <- model.matrix(winner ~ ., data = election.te)[, -1]
y_test <- election.te$winner

y_train_num <- ifelse(y_train == "gop", 1, 0)
y_test_num <- ifelse(y_test == "gop", 1, 0)

lambda_grid <- seq(1, 50) * 1e-4

set.seed(20)
```

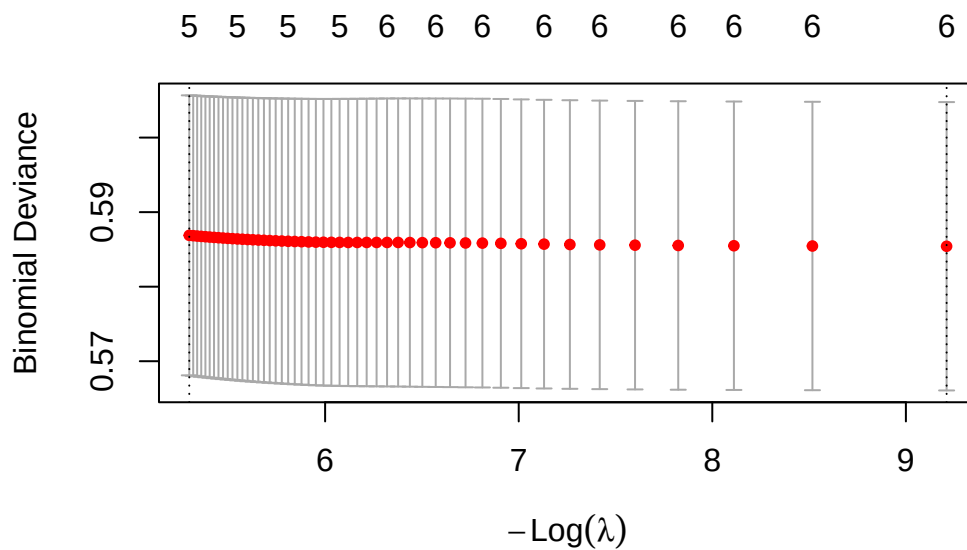


```
lasso_cv <- cv.glmnet(
  x = x_train,
  y = y_train_num,
  family = "binomial",
  alpha = 1,
  lambda = lambda_grid,
  nfolds = 10
)

best_lambda <- lasso_cv$lambda.min
best_lambda
```

```
[1] 1e-04
```

```
plot(lasso_cv)
```



```
lasso_coef <- coef(lasso_cv, s = "lambda.min")
lasso_coef
```

```
7 x 1 sparse Matrix of class "dgCMatrix"
      lambda.min
```

```

(Intercept)          -6.490197e+00
Urban1               -1.422961e+00
Median_Household_Income -1.138489e-08
pct_male             6.697114e+00
pct_whitealone       7.804548e+00
pct_black            7.112388e-01
pct_hispanic         1.593396e+00

```

```

nonzero_coef <- lasso_coef[lasso_coef[, 1] != 0, , drop = FALSE]
nonzero_coef

```

7 x 1 sparse Matrix of class "dgCMatrix"

```

              lambda.min
(Intercept)          -6.490197e+00
Urban1               -1.422961e+00
Median_Household_Income -1.138489e-08
pct_male             6.697114e+00
pct_whitealone       7.804548e+00
pct_black            7.112388e-01
pct_hispanic         1.593396e+00

```

```

lasso_train_prob <- predict(lasso_cv, newx = x_train, s = "lambda.min", type = "response")
lasso_test_prob <- predict(lasso_cv, newx = x_test, s = "lambda.min", type = "response")

```

```

lasso_train_pred <- ifelse(lasso_train_prob > 0.5, "gop", "dem")
lasso_test_pred <- ifelse(lasso_test_prob > 0.5, "gop", "dem")

```

```

lasso_train_pred <- factor(lasso_train_pred, levels = levels(y_train))
lasso_test_pred <- factor(lasso_test_pred, levels = levels(y_test))

```

```

records["lasso", "train.error"] <- mean(lasso_train_pred != y_train)
records["lasso", "test.error"] <- mean(lasso_test_pred != y_test)

```

```
records
```

	train.error	test.error
tree	0.1004419	0.1171750
logistic	0.1004419	0.1155698
lasso	0.1145038	0.1396469

The optimal lambda from cross validation was 0.0001. All predictors remained in the LASSO model so none were removed. This suggests each variable contributed useful information for

predicting the winning candidate. Logistic regression had the lowest test error (13.96%). Overall logistic regression performed the best and regularization did not improve accuracy for this dataset.

16. Compute ROC Curves

```
library(pROC)
```

Warning: package 'pROC' was built under R version 4.5.2

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

cov, smooth, var

```
tree_prob <- predict(pruned_tree, newdata = election.te, type = "vector")[, "gop"]

log_prob <- predict(log_fit,
                    newdata = election.te,
                    type = "response")

lasso_prob <- predict(lasso_cv,
                     newx = x_test,
                     s = "lambda.min",
                     type = "response")

y_true <- ifelse(election.te$winner == "gop", 1, 0)

roc_tree <- roc(y_true, as.numeric(tree_prob))
```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

```
roc_log <-roc(y_true, as.numeric(log_prob))
```

Setting levels: control = 0, case = 1

Setting direction: controls < cases

```
roc_lasso <-roc(y_true, as.numeric(lasso_prob))
```

Setting levels: control = 0, case = 1

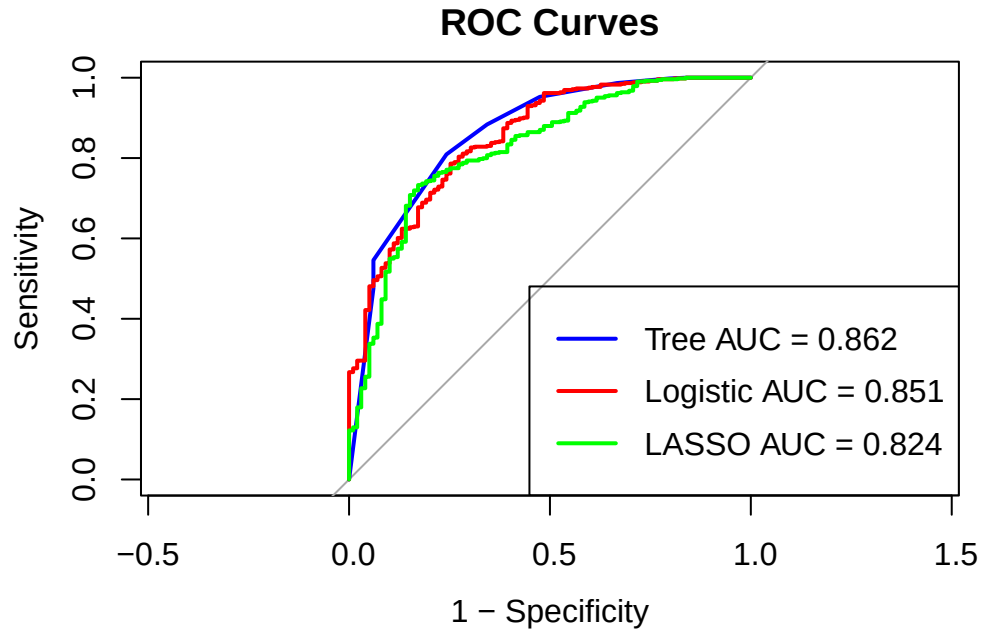
Setting direction: controls < cases

```
plot(roc_tree,
     col = "blue",
     main = "ROC Curves",
     legacy.axes = TRUE)

plot(roc_log,
     col = "red",
     add = TRUE)

plot(roc_lasso,
     col = "green",
     add = TRUE)

legend("bottomright",
     legend = c(
       paste("Tree AUC =", round(auc(roc_tree),3)),
       paste("Logistic AUC =", round(auc(roc_log),3)),
       paste("LASSO AUC =", round(auc(roc_lasso),3))
     ),
     col = c("blue","red","green"),
     lwd = 2)
```



```
auc(roc_tree)
```

Area under the curve: 0.862

```
auc(roc_log)
```

Area under the curve: 0.8508

```
auc(roc_lasso)
```

Area under the curve: 0.824

The ROC curves show that all three models perform well at predicting election outcomes. The Decision Tree had the highest AUC (0.862) then followed by Logistic AUC (0.851) and LASSO AUC (0.824). This indicated that the decision tree provided the best classification performance on the test data. However logistic regression and LASSO offer greater interpretability. With LASSO also helping reduce overfitting through regularization. Overall the decision tree was the best for prediction while the regression based methods were better for understanding the effects of individual variables.

Taking it Further

17. Explore additional methods

We will try linear discriminant analysis and quadratic discriminant analysis.

LDA:

Fitting:

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

```
select
```

```
lda_fit <- lda(winner ~ ., data = election.tr )
```

Prediction:

```
lda <- c(0, 0)
records <- rbind(records, lda)
lda_train_pred <- predict(lda_fit, election.tr)$class
lda_test_pred <- predict(lda_fit, election.te)$class
records["lda", "train.error"] <- mean(lda_train_pred != election.tr$winner)

records["lda", "test.error"] <- mean(lda_test_pred != election.te$winner)
```

LDA and Lasso are weaker than tree and logistic regression in error. This weakness suggests that the data may not fully fulfill the LDA assumptions of normal predictors and equal covariance between predictors.

QDA:

Fitting:

```
qda_fit <- qda( winner ~ ., data = election.tr )
```

Pred:

```

qda <- c(0, 0)
records <- rbind(records, qda)
qda_train_pred <- predict(qda_fit, election.tr)$class
qda_test_pred <- predict(qda_fit, election.te)$class
records["qda", "train.error"] <- mean(qda_train_pred != election.tr$winner)
records["qda", "test.error"] <- mean(qda_test_pred != election.te$winner)
records

```

	train.error	test.error
tree	0.1004419	0.1171750
logistic	0.1004419	0.1155698
lasso	0.1145038	0.1396469
lda	0.1137003	0.1380417
qda	0.6801928	0.6725522

QDA has much higher error than the others. It is not finding a useful structure to interpret the data. It may be because some features are probably correlated, White and MHI, for example.

18. We will try linear regression to predict candidate votes.

```

census_raw <- read.csv("census.csv")
census_lr <- census_raw %>%
  filter(State != "Puerto Rico")
lr_data <- election_raw %>%
  left_join(census_lr, by = c("County", "State"))
lr_data <- na.omit(lr_data)

```

fitting:

```

model_lr <- lm(votes_gop ~ Total.Population + White.Alone + Black.or.African.American.Alone +
summary(model_lr)

```

Call:

```

lm(formula = votes_gop ~ Total.Population + White.Alone + Black.or.African.American.Alone +
  Hispanic.or.Latino + Median.Household.Income + Urban.Rural,
  data = lr_data)

```

Residuals:

Min	1Q	Median	3Q	Max
-201161	-1584	-398	897	165308

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.037e+03	3.262e+02	3.177	0.00150	**
Total.Population	-7.475e-02	6.441e-03	-11.607	< 2e-16	***
White.Alone	3.999e-01	7.931e-03	50.420	< 2e-16	***
Black.or.African.American.Alone	3.060e-02	8.883e-03	3.444	0.00058	***
Hispanic.or.Latino	1.232e-01	6.800e-03	18.113	< 2e-16	***
Median.Household.Income	1.451e-06	1.360e-05	0.107	0.91505	
Urban.RuralUrban	3.564e+03	5.059e+02	7.045	2.28e-12	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12810 on 3105 degrees of freedom

Multiple R-squared: 0.9491, Adjusted R-squared: 0.949

F-statistic: 9658 on 6 and 3105 DF, p-value: < 2.2e-16

I fitted raw totals from the census data onto the raw GOP vote totals for each county. I used total population, the three races, median household income, and the Urban/Rural categorical variable. This information is not very helpful, since it makes sense that total GOP vote will go up if any of the raw vote totals go up, especially for urban areas, which is what we see here. What is interesting is that for a 100 person increase in total population, gop vote actually goes down by 7.458 persons, which shows the GOP's weakness in higher-population counties.

19. Summary

Demographics: Race plays a significant role in partisan politics at the presidential level. The percentage of the white population was the top predictor in the decision tree and our regressions showed that a 1% increase in white population increased the odds of a county going republican by almost 8%.

We were able to get nuanced analysis from the tree model, like how majority nonwhite counties actually leaned GOP if they were sufficiently poor, showing the GOP's gains among lower income people. We were able to connect our own California trends to the tree model; which is that wealthy counties in the US with a sub-70% white population lean democratic, a phenomenon that is widespread in California.

The Linear regression model highlighted the urban-rural divide by showing that for every 100-person increase in total population in a county, raw GOP vote actually decreases by 7.5 votes, meaning they are hemorrhaging votes in larger counties with large cities.

Model comparison: the decision tree was the best at classification since it had the highest AUC, though logistic regression had the lowest test error and is better for interpretation, as

demonstrated above. The Lasso model showed that all of the selected census variables were useful to classifying a winner, since none of them were forced to zero by the lambda. We know that there are strong correlations between at least some of these census features, since the QDA didn't work properly (it assumes no significant correlations between features). PCA was able to place Santa Barbara County into a cluster of 15 similar counties, meaning it was effective in reducing noise from the raw data.

Issues: The linear regression was problematic since we used raw totals, which are skewed by overall county size, meaning our coefficients were not very meaningful. In the future, we could transform all the raw totals, including democratic and republican totals, into percentages.

Plenty of additional data could also make our models more robust. If we add education or religion, perhaps they are more meaningful than our existing predictors and the LASSO would set some of our existing predictors to zero after adding them.